# Demand for Labour in Competitive Labour Markets

ECON 300 - Labour Economics

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# Learning Objectives

- Understand how firms decide how much labour to employ to produce a given level of output.
- Distinguish **short run** (some inputs fixed) from **long run** (all inputs variable).
- See why labour demand is **downward sloping** with respect to the wage.
- Learn the determinants of the **elasticity of labour demand**.
- Analyze impacts of **free trade**, globalization, offshoring, and ICT on labour demand.
- Compare implications under **perfect competition** vs. **monopoly** in product markets.

# Main Questions

- Is there empirical support for downward-sloping labour demand?
- How competitive is Canadian labour globally? How have productivity and wages evolved?
- What is the impact of the Internet and technological change on labour demand?
- How can we use labour demand to assess offshoring/globalization effects on wages and employment?

# Derived Demand and Time Horizons

- **Derived demand:** Demand for labour depends on demand for the goods/services it produces.
- **Short run:** At least one input (e.g., capital) is fixed.
- **Long run:** All inputs are variable; firms can adjust capital and labour.

# Objectives, Constraints, and Decisions

**Objective:** Profit maximization.

- Constraints: market structure, product demand, factor prices, production function, time frame.
- Labour demand: desired quantity of labour given the wage or a labour supply schedule faced by the firm.

# Product & Labour Market Structures

- Output and labour market structures are **independent**.
- Together they shape wage and employment outcomes via demand and supply interactions.

# Production Function and Assumptions

- Two inputs: Labour $N$ and capital $K$; output $Q = F(K, N)$.
- In the short run, $K$ is fixed; only $N$ varies.

# Profit-Maximization: Rules of Thumb

- **Operate** if total revenue covers total *variable* cost.
- Expand output until MC $=$ MR.

# Revenue Concepts for Input Decisions

- **TRP** (Total Revenue Product): revenue from employing a given amount of an input.
- **MRP** (Marginal Revenue Product): change in total revenue from a marginal unit of the input.

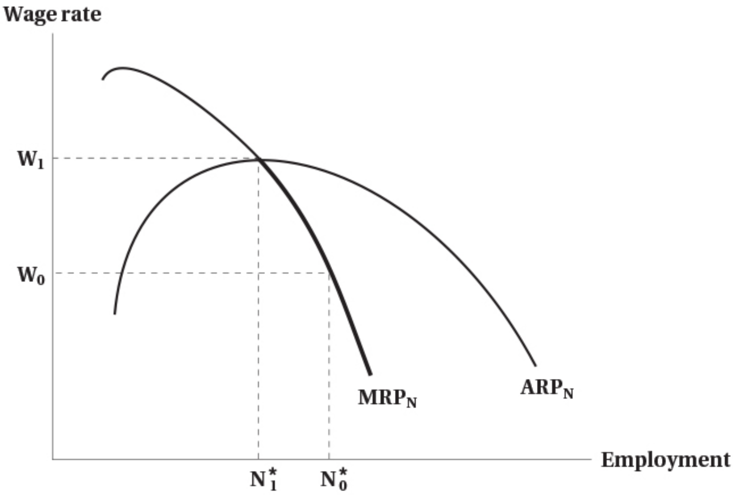
# Short-Run Hiring Rule

- Produce if $\text{TRP}_N > \text{TC}_N$.
- Hire labour until $\text{MRP}_N = \text{MC}_N = w$.

# Competitive Firm: Properties

- Price taker in output and labour markets.
- Can hire labour without affecting the market wage; $\text{MC} = \text{AC} = w$.
- Hire until $\text{MRP} = w$; the **short-run labour demand** is the $\text{MRP}_N$ curve.

# Figure 5.1: The Firm's Short-Run Demand for Labour



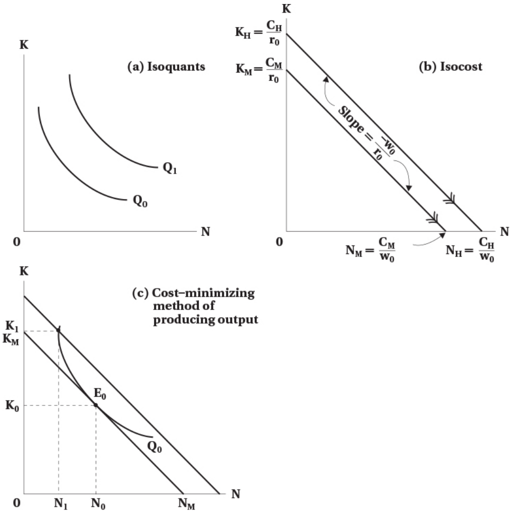
# Shutdown Region and Curve Segment

- Shut down if the wage exceeds the **average revenue product** of labour.
- Short-run labour demand coincides with $\text{MRP}_N$ *below* the $ARP$–$MRP$ intersection.
- **Downward sloping:** Diminishing marginal returns; $\downarrow w \Rightarrow \uparrow N$.

# Isoquants and Isocosts

- **Isoquant:** Combinations of $K$ and $N$ yielding a given $Q$; diminishing MRTS.
  - **MRTS** – the rate at which a firm can substitute labor for capital while keeping production output constant
- **Isocost:** $TC = rK + wN$, all input bundles with the same cost.

# Figure 5.2: Isoquants, Isocost, and Cost Minimization



# Panel (a): Isoquants

- Curves labeled $Q_0$ and $Q_1$ are **isoquants**
- Each isoquant shows all $(K, N)$ combinations producing the same output
- $Q_1$ represents a higher level of output than $Q_0$
- Isoquants are:
  - Downward sloping (inputs are substitutes)
  - Convex to the origin (diminishing MRTS)

# Isoquants: Interpretation

- Moving **along** an isoquant:
  - Output remains fixed
  - Firm substitutes capital for labour or vice versa
- Moving to a **higher isoquant** (from $Q_0$ to $Q_1$):
  - Requires more inputs
  - Represents an increase in output
- Technological change (not shown):
  - Inward shift: productivity improvement
  - Outward shift: productivity decline

# Panel (b): Isocost Lines

- Isocosts show combinations of $K$ and $N$ with the same total cost
- Total cost:

$$C = w_0 N + r_0 K$$

- $w_0$: wage rate, $r_0$: rental rate of capital
- Lines labeled $C_M$ and $C_H$ represent different cost levels

# Isocost Lines: Slopes and Intercepts

- Vertical intercepts:

$$K_M = \frac{C_M}{r_0}, \qquad K_H = \frac{C_H}{r_0}$$

- Horizontal intercepts:

$$N_M = \frac{C_M}{w_0}, \qquad N_H = \frac{C_H}{w_0}$$

- Slope of the isocost:

$$\text{Slope} = -\frac{w_0}{r_0}$$

- Interpretation: trade-off between labour and capital holding cost fixed

# Shifts and Rotations of Isocosts

- Increase in total cost $C$:
  - Parallel outward shift of the isocost
- Increase in wage $w_0$:
  - Isocost becomes steeper
  - Rotates inward around the $N$-intercept
- Increase in rental rate $r_0$:
  - Isocost becomes flatter
  - Rotates inward around the $K$-intercept

# Panel (c): Cost-Minimizing Choice

- Firm wants to produce $Q_0$ at the lowest possible cost
- Chooses point $E_0$ where:
  - Isoquant $Q_0$ is tangent to the lowest isocost
- Optimal input bundle:

$$(K_0, N_0)$$

- Any other point on $Q_0$ lies on a higher-cost isocost

# Tangency Condition at the Optimum

- At the cost-minimizing point $E_0$:

$$MRTS_{N,K} = \frac{MPP_N}{MPP_K}$$

- Cost minimization requires:

$$\frac{MPP_N}{MPP_K} = \frac{w_0}{r_0}$$

- Interpretation:
  - The rate at which the firm can trade $K$ for $N$ in production
  - Equals the rate at which the market allows substitution via prices

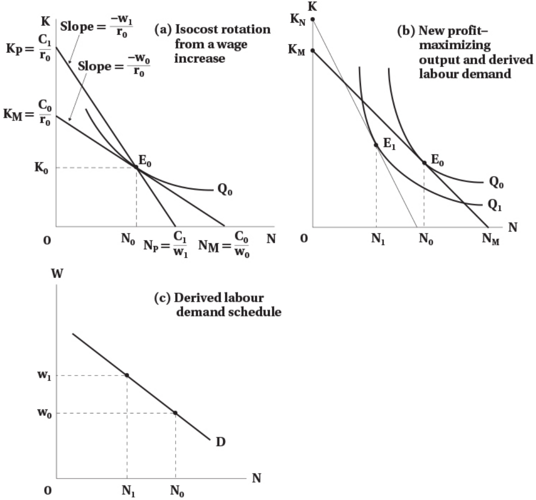
# Why Other Points Are Not Optimal

- Points like $(K_1, N_1)$:
    - Produce $Q_0$ but lie on higher isocosts
    - Are more expensive than $E_0$
- Points closer to the origin but not tangent:
    - Either infeasible or violate cost minimization
- Only tangency ensures minimum cost for given output

# Movements vs Shifts

- **Movement along an isoquant**:
  - Output fixed
  - Input mix changes
  - Driven by changes in relative prices
- **Shift of isoquants**:
  - Caused by technological change
  - Affects productivity and labour demand
- **Rotation of isocosts**:
  - Caused by changes in $w_0$ or $r_0$
  - Drives substitution between labour and capital

# Figure 5.3: Deriving the Labour Demand Schedule



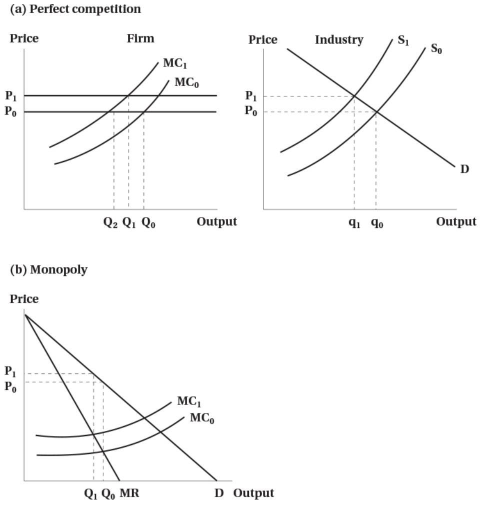
# Cost Increases Under Perfect Competition

- Higher $w$ rotates the isocost inward; substitute $K$ for $N$.
- Firm MC and AC shift up; industry output falls and price rises.

# Monopoly Case: Cost Increases

- Monopolist: $MR = MC$, price setter, $P > MC$.
- Hiring more labour lowers both MPP and MR; output and employment responses differ from competition.

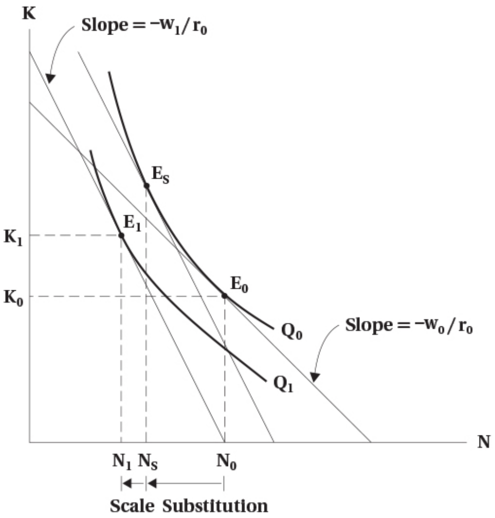
# Figure 5.4: The Effect of a Cost Increase on Output



# Substitution vs. Scale Effects

- **Substitution effect:** Use cheaper inputs instead of labour.
- **Scale effect:** Higher costs shrink output scale, reducing all inputs.

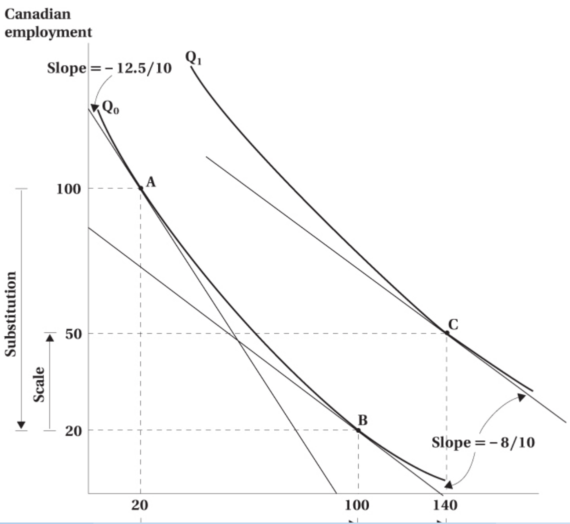
# Figure 5.5: Substitution and Scale Effects of a Wage Change



# Worked Example: Offshoring (Toy Inc.)

Toy Inc. can hire assembly workers in Canada or abroad; workers are **close substitutes**. Outsourcing abroad $\Rightarrow$ lower effective wage for that task $\Rightarrow$ **reduces Canadian employment**.

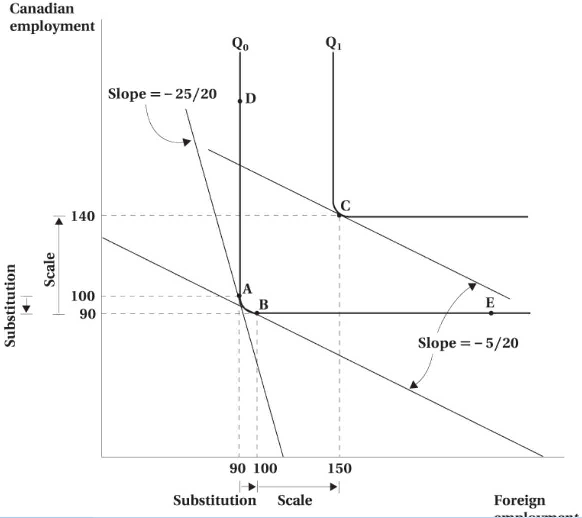
# Toy Inc.


Canadian employment
Slope = − 12.5/10
$Q_1$
$Q_0$
A
100
C
50
Substitution
Scale
B
20
Slope = − 8/10
20
100
140

# Worked Example: Offshoring (Bridge Inc.)

Bridge Inc. builds bridges; Canadian engineers/designers/managers work at home; construction workers hired abroad. Inputs are **complements**. Outsourcing on-site construction **does not reduce** Canadian employment of complementary skilled workers.

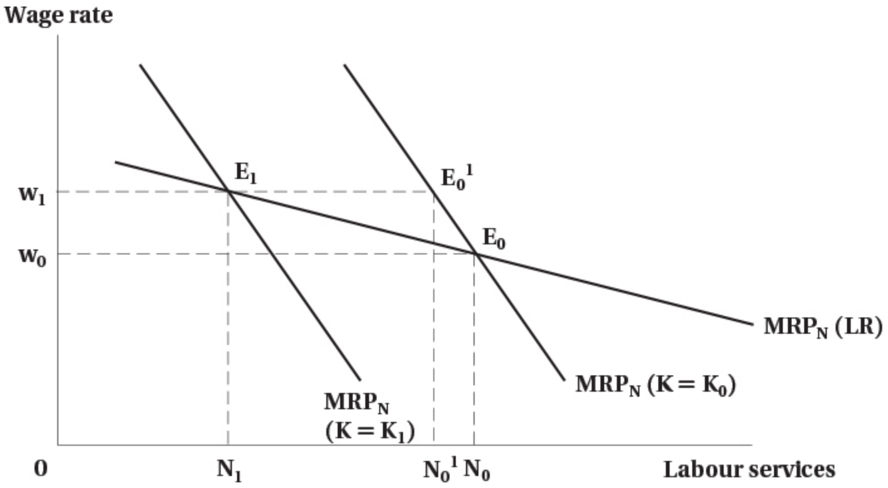
# Bridge Inc. — Illustration



# Linking Horizons

- **Short run:** $K$ fixed; no substitution effect.
- **Long run:** $K$ variable; substitution $\Rightarrow$ larger employment response to wage changes.

# Figure 5.6: Labour Demand in Short vs. Long Run



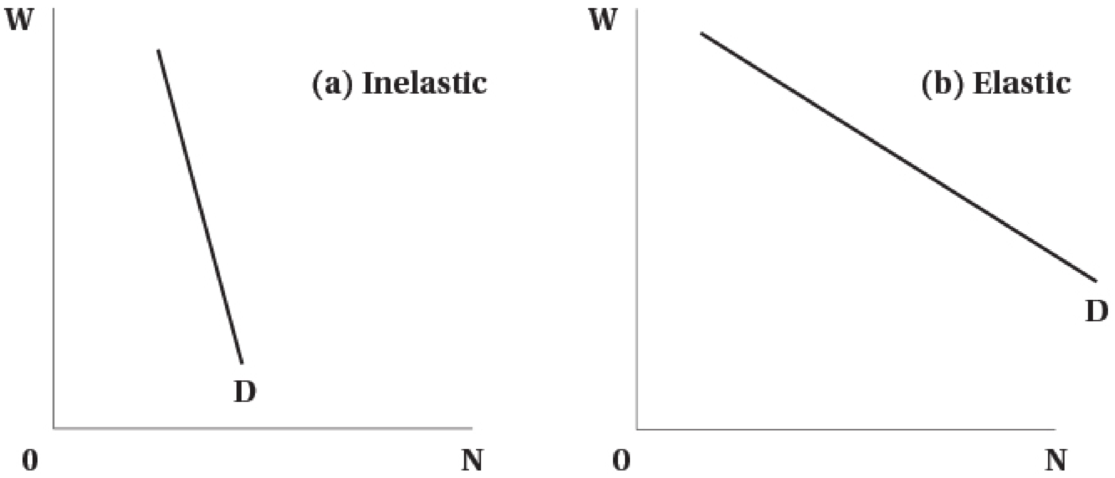
# Cost Minimization in Public/Quasi-Public Sectors

Organizations target **given output at minimum cost**. Scale vs. substitution effects remain key to labour demand responses.

# Definition and Measurement

- Demand for labour typically falls when wages rise.
- Elasticity $\varepsilon_{N,w} = \frac{\%\ \Delta N}{\%\ \Delta w}$.

# Figure 5.7: Inelastic vs. Elastic Labour Demand



# Determinants of Elasticity

- Availability of **substitute inputs**.
- Supply conditions of substitute inputs.
- Elasticity of demand for **output**.
- Ratio of **labour cost** to **total cost**.

# Implications

- Poor substitutability $\Rightarrow$ inelastic labour demand.
- Inelastic product demand $\Rightarrow$ inelastic labour demand.
- Small labour-cost share $\Rightarrow$ inelastic labour demand.

# Changing Demand Conditions

- TPP and other trade agreements.
- Offshoring and ICT diffusion.
- Media narratives vs. measured employment impacts.

# The Impact of Trade in a Single Labour Market

- Production does not automatically move to lower-wage countries.
- Profit-maximization depends on **both** costs and productivity.
- Foreign and Canadian labour may be **imperfect substitutes**.

# TABLE 5.1: Productivity (2000) and Changes in Productivity, Hourly Compensation, and Unit Labour Costs, 2000–2018

| Country | Productivity in 2000 (as % of Canada) | % Change in Productivity | % Change in Compensation |
|---|---|---|---|
| Korea | 45 | 99 | 32 |
| Mexico | 46 | 2 | –44 |
| New Zealand | 79 | 21 | 76 |
| Japan | 87 | 20 | –38 |
| Australia | 99 | 24 | 45 |
| Canada | 100 | 18 | 16 |
| Spain | 102 | 17 | 18 |
| United Kingdom | 111 | 18 | –5 |
| Finland | 116 | 20 | 24 |
| Italy | 119 | 1 | 14 |
| Sweden | 121 | 28 | 17 |
| United States | 123 | 30 | 2 |
| Germany | 126 | 19 | 9 |
| France | 129 | 18 | 19 |
| Netherlands | 135 | 15 | 19 |
| Denmark | 135 | 22 | 33 |
| Switzerland | 136 | 19 | 40 |
| Belgium | 143 | 14 | 21 |
| Norway | 164 | 16 | 46 |

# Figure 5.8: Relative Trends in Labour Costs and Productivity (Canada vs. U.S.)

# Summary

- **Derived demand:** Labour demand stems from product demand.
- **Short run:** Hire to MRP $= w$.
- **Long run:** Substitution and scale effects; demand more elastic.
- **Elasticity drivers:** Substitutability, output demand elasticity, labour-cost share.
- **Globalization/ICT:** Channels run via productivity, costs, and task (substitute vs. complement).

# Thank You!